

Can control barrier functions keep automated vehicles safe in live freeway traffic?

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ABSTRACT

This work presents testing of a Control Barrier Function (CBF) supervised Automated Vehicle (AV) in live freeway traffic. The CBF is designed and implemented using a common dynamical model for AV longitudinal control and a time-gap based safety property combined with CAN bus injection software/hardware. 1.17 hours of car-following data is collected from driving in congested freeway traffic. We analyze the extent to which the CBF controlled AV satisfied three properties: 1) forward-invariance of the safety property, 2) recovery of the safety property, and 3) collision avoidance.

Our main findings are as follows. Forward-invariance was not strictly satisfied across all states. When trajectories begin satisfying the safety property it was violated by a maximum of $3.4[m]$, and 90% of violations were less than $2[m]$. Recovery was also not strictly satisfied across all states. For trajectories which begin outside of the safe set due to merge-in events, if the violation to the safety property was by more than $5[m]$ the AV was recovering back to the safe set in more than 98% of the time. Finally, the minimum spacing-gap was $11.7[m]$. Across the tests the AV remained far from any collisions. We additionally analyze errors between control inputs and achieved outputs and hypothesize that unaccounted for modeling errors may lead to under-braking compared to what the CBF logic specifies, which may contribute to the observed property violations.

CCS CONCEPTS

• **Computer systems organization** → **Robotic autonomy**; **Robotic control**; **Sensors and actuators**; *Reliability*; • **Applied computing** → *Engineering*.

KEYWORDS

Control barrier functions, safe autonomy, automated vehicles, transportation

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1 INTRODUCTION

A recently emerging approach for formally guaranteed safety of autonomously controlled cyber-physical systems is active safety filtering (ASF) [18, 28]. An ASF provides sets of safe control inputs defined over systems states, often called the safety kernel or the viability kernel [3, 6], such that if control is continually chosen from within the set of safe control inputs at all times the system has a formal guarantee of forward invariance in safety. This is an attractive approach for formal safety in that it allows otherwise potentially unsafe controllers to be supervised for safety. Common approaches for ASF are reach-set analysis, model-predictive control, and control barrier functions.

Control barrier functions (CBFs) have received significant attention in recent years [3, 5, 30]. By relating Lie derivatives of safety properties to control inputs CBFs can be developed to provide analytical representations of safety kernels. The ensuing safety kernel can then be used potentially in real-time to perform safety supervision of potentially unsafe nominal control inputs. CBFs have been shown to have a number of desirable theoretical properties, such robustness to disturbances [10, 17, 33], ability to incorporate data-driven approaches [26, 27, 32], and applicability to multi-agent systems [9, 29].

Of particular importance for this work, CBFs are frequently shown to be readily applicable to the control of AVs. Initial demonstrations of the value of CBFs showed how Adaptive Cruise Controllers could be synthesized [4, 5, 33]. The development of CBFs for general safety supervision of AVs is especially relevant to research in transportation given the current interest in using AVs as control mechanisms to influence broader traffic flows with (potentially unsafe) control algorithms [12, 13, 19, 25].

While the development of theory for CBFs has advanced quickly, there is a significant need for more extensive testing and validation. In [2] a CBF was fielded onto an automated truck and deployed to a closed testing track, where it was found to achieve safety. The work in [1] extends this to a deployment involving connectivity across vehicles. This work expands on these experimental efforts to provide real world demonstrations of the performance of CBFs for safe control of AVs. We deploy an AV with safety supervised by a CBF into a live freeway environment, rather than on a closed track.

The need for experimental demonstrations of technologies related to automated vehicles in traffic flow is well documented.



In [25] experimental demonstration of the capacity for an AV to control a broader traffic flow motivated significant further research after initial theoretical demonstrations demonstrated the possibility. Alternatively, in several works [15, 16, 20] experimental testing of commercially available adaptive cruise control vehicles demonstrated a gap between theoretical demonstrations of the capacity to positively benefit traffic flows and their actual performance, which underperformed compared to theoretical guarantees. In [11] the mismatch between theoretical performance of automated vehicles in traffic flows was further documented. The work in [22, 23] gave experimental demonstrations of both the difficulties and possibilities of using vehicle to infrastructure communication for automated compliance with variable speed limits. Control barrier functions, and other active safety filtering approaches, have received significant attention theoretically, but there is a lack of experimental deployment and testing in full-scale contexts. For this work we provide similar live experimental results of how a common CBF formulation for safe control of AVs actually performs in live traffic.

Contributions. The primary contribution of this work is to experimentally test the performance of a control barrier function supervised automated vehicle in live traffic. We collect and analyze 1.2 hours of car-following under CBF supervision on a live freeway. The CBFs control performance is assessed in terms of three properties: forward invariance of a minimum time-gap safety property, recovery of that safety property if violated, and forward invariance of collision-avoidance. We find that the first two properties are not strictly met in experimentation due to modeling error. We also find the degree of violation of those properties is bounded and small. No collisions were observed. This work is the first to provide experimental testing of a CBF controlled AV in a live freeway environment.

Organization. The organization of the work is as follows. In Section 2 the mathematical background on control barrier functions and control of automated vehicles is provided. Section 3 gives details on the experimental testing performed. Results of the experiments are provided in 4. In 5 we conclude the work and discuss future research directions.

2 BACKGROUND

This section provides a brief mathematical background on control barrier functions. First we give the general conditions needed for forming and using a CBF. We expand the standard CBF formulation to encompass environmental CBFs. Finally, we give conditions for CBFs to recover a system back to safety in the case of a safety violation.

2.1 Control barrier functions as active safety filters

The work in [30] introduces control barrier functions, and in [3, 5] this was expanded to allow for supervision of potentially unsafe control algorithms. Here we give an overview of the CBF formulation.

Consider control systems of the following form:

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t)), \quad (1)$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ is state of the system at time t , $\mathbf{u}(t) \in \mathbb{R}^m$ is the control input to the system, and $f(\cdot, \cdot) : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$, are the continuously differentiable system dynamics.

Let $h(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a safety quantity defined on the system state. Given $h(\mathbf{x})$, the system's safety is defined as a property as follows:

$$h(\mathbf{x}) \geq 0, \quad (2)$$

where safety is satisfied or not (e.g., true or false) at a given state $\mathbf{x}$. Additionally, let $\mathcal{S} = \{\mathbf{x} : h(\mathbf{x}) \geq 0\}$ be the set of safe states.

Forward-invariance in $\mathcal{S}$ can be stated as:

$$h(\mathbf{x}(t_0)) \geq 0 \implies h(\mathbf{x}(t)) \geq 0, \forall t > t_0. \quad (3)$$

where t_0 is initial time.

A given h is a CBF if a function $\alpha(\cdot)$ exists in extended class- $\mathcal{K}_{\text{inf}}$ such that:

$$\sup_{\mathbf{u}} \dot{h}(\mathbf{x}, \mathbf{u}) + \alpha(h(\mathbf{x})) \geq 0, \forall \mathbf{x} \in \mathcal{S}, \quad (4)$$

where $\dot{h}$ is the time derivative of h along (1), with:

$$\dot{h}(\mathbf{x}, \mathbf{u}) = \langle \nabla h(\mathbf{x}), f(\mathbf{x}, \mathbf{u}) \rangle.$$

A CBF can be used to create an *active safety filter* for choosing control inputs that will render the system in (1) forward invariant with respect.

The pointwise set of the control inputs ($\mathbf{u}$) that, at a given $\mathbf{x}$, will satisfy (4) are:

$$\mathcal{U}_{\text{safe}}(\mathbf{x}) = \{\mathbf{u} : \langle \nabla h(\mathbf{x}), f(\mathbf{x}, \mathbf{u}) \rangle + \alpha(h(\mathbf{x})) \geq 0\}. \quad (5)$$

$\mathcal{U}_{\text{safe}}(\mathbf{x}) \subset \mathcal{R}^m$ is frequently referred to as the *safety kernel*. Any Lipschitz continuous controller $u = k(\mathbf{x})$ such that $k(\mathbf{x}) \in \mathcal{U}_{\text{safe}}(\mathbf{x}), \forall \mathbf{x} \in \mathcal{S}$ will render (1) forward-invariant with respect to a given safety property.

CBFs are often used to supervise another nominal controller $k_{\text{nom}}(\mathbf{x})$ which may not have safety guarantees. A CBF can be used as an active safety filter on k_{nom} by overriding control inputs that are not inside $\mathcal{U}_{\text{safe}}(\mathbf{x})$ and choosing a control input that is. This is often framed as solving the following optimization problem:

$$\begin{aligned} \min_{\mathbf{u}} \quad & \|u - k_{\text{nom}}(\mathbf{x})\| \\ \text{s.t.} \quad & u \in \mathcal{U}_{\text{safe}}(\mathbf{x}) \end{aligned} \quad (6)$$

which chooses the u closest to the desired nominal control that is inside $\mathcal{U}_{\text{safe}}(\mathbf{x})$. In this work a nominal controller designed to do speed control is supervised in this manner.

2.2 Environmental CBFs

Environmental CBFs [21] (E-CBFs) are CBFs that consider safety properties that depend on measurements from outside of the system dynamics. For instance, in this work the AV responds to the driving dynamics of leading vehicles. These dynamics are considered outside of the control of the AV. Additionally, the environmental dynamics are not known via a model and must be sensed. We include this framing of the problem to emphasize that the success of the CBF in this case depends on reacting to an outside live dynamical environment, rather than a closed dynamical environment as is assumed in many works.

Let $\mathbf{e} \in \mathbb{R}^l$ be the state of an outside environment. We rewrite (2) as:

$$h(\mathbf{x}, \mathbf{e}) \geq 0.$$

with $h(\mathbf{x}, \mathbf{e}) : \mathcal{R}^n \times \mathcal{R}^l \rightarrow \mathcal{R}^1$ a safety property that considers both the state of the controlled dynamical system and the environmental state.

Rewriting the safety kernel definition from (5) to account for environmental signals yields:

$$\mathcal{U}_{\text{safe}}(\mathbf{x}, \mathbf{e}, \dot{\mathbf{e}}) = \{\mathbf{u} : \dot{h}(\mathbf{x}, \mathbf{u}, \mathbf{e}, \dot{\mathbf{e}}) + \alpha(h(\mathbf{x}, \mathbf{e})) \geq 0\}. \quad (7)$$

which depends not just on $\mathbf{x}$, but also $\mathbf{e}$ and $\dot{\mathbf{e}}$. As the outside environmental state changes the safety kernel changes with respect to it. Again, in the context of safe supervision of an AV in live traffic this corresponds to a safety kernel that is formed with respect to the state of other vehicles in the traffic flow environment.

2.3 Safety recovery

Safety recovery for CBFs was introduced in [33] as the following property:

$$h < 0 \implies \dot{h} > 0. \quad (8)$$

meaning that any time the safety property is violated the the safety property is also strictly increasing. If the system is always recovering when outside of the safe set it will eventually return to within safety.

In the context of an AV driving in traffic violations to the safety property may happen due to vehicles in other lanes merging in front of the AV at distances that violate the safety property. Recovery for the AV in this context means returning back to a safe following distance.

Even for $\mathbf{x}$ outside $\mathcal{S}$ the safety kernel may still defined and non-empty. In this case, the following holds:

$$\mathbf{u} \in \mathcal{U}_{\text{safe}}(\mathbf{x}) \forall \mathbf{x} \notin \mathcal{S} \implies \dot{h} > 0$$

which states that if control inputs are still chosen from the safety kernel when outside of the safe set, then the state of the system will return back to the safe set. This is because if $\dot{h}$ is strictly increasing when $h < 0$, eventually the state will return to $h \geq 0$. This can be seen from the following:

$$h < 0 \wedge \dot{h} + \alpha(h) \geq 0 \implies \dot{h} > 0, \quad (9)$$

as $h < 0 \implies \alpha(h) < 0$.

In the context of control of an AV, recovery refers to the AV moving back to a following distance that satisfies the safety property.

2.4 CBFs for safe car-following with AV control

Here we provide a description of the dynamical model for controlling an *automated vehicle* (AV) in the context of car-following. Additionally, the E-CBF used in the remainder of the work is presented.

Let the dynamics of a single AV to be controlled be as follows:

$$\dot{\mathbf{x}}(t) = \begin{pmatrix} \dot{p}(t) \\ \dot{v}(t) \end{pmatrix} = \begin{pmatrix} v(t) \\ u(t) \end{pmatrix}, \quad (10)$$

where $p(t)$ is vehicular position, $v(t)$ is vehicular speed, and $u(t)$ is the commanded vehicular acceleration. In this work, the modeling assumption made for control is that vehicular acceleration can be controlled directly.

The measured signals ($\mathbf{e}(t)$) are as follows:

$$\mathbf{e}(t) = \begin{pmatrix} s(t) \\ \Delta v(t) \end{pmatrix}, \quad (11)$$

where $s(t)$ is the measured spacing/positional difference, and $\Delta v(t)$ is the measured speed difference. Both are measured by on-board radar/camera.

A common desired safety property is to maintain a minimum time gap to the vehicle ahead. This is written as:

$$h(\mathbf{x}, \mathbf{e}) = s - (t_{\min}v + d_{\min}), \quad (12)$$

where $t_{\min}[s]$ is a the minimum following time-gap, and $d_{\min}[m]$ is the minimum following distance, each of which can be chosen to adjust driving behavior. Forward invariance for this property can be stated in terms of state of the AV as follows:

$$s(t_0) \geq (t_{\min}v(t_0) + d_{\min}) \implies s(t) \geq (t_{\min}v(t) + d_{\min}) \forall t \geq t_0 \quad (13)$$

where t_0 is an initial time.

The time-derivative of h given the model in (10) is as follows:

$$\dot{h}(\mathbf{x}, \mathbf{u}, \mathbf{e}, \dot{\mathbf{e}}) = \Delta v - t_{\min}u. \quad (14)$$

which is a function of u , given the modeling assumption that $\dot{v} = u$.

Recovery for the minimum time-gap property can be stated as the following property:

$$s < (t_{\min}v + d_{\min}) \implies \Delta v + t_{\min}u > 0 \quad (15)$$

3 EXPERIMENTAL DETAILS

Here details of the experimental data collection are provided. First, we provide details on the hardware and software platform used to perform control and data collection. Next, We give the structure of the control algorithms used on the AV. We describe processing that is done on the measured data. Finally, we describe prevailing traffic conditions in which the experiment is conducted.

The automated vehicle used in experiments is a 2021 Toyota Rav-4 retrofitted with CAN bus read/write hardware and software. Driving was conducted along Interstate 24 in the United States near Nashville, TN. Data was collected between the hours of 6 AM and 8 AM CST, which correspond to morning rush hours along this corridor.

Hardware/software. We use a low-cost and reversible instrumentation to extend a commodity vehicle into one which hosts experimental automated control. The core hardware components are: (1) a vehicle-specific wiring harness, attached to (2) a printed circuit board with CAN transceivers and relay switch for CAN transmissions, and (3) an embedded computer hosting key software for data and control. Libpanda [7] manages the data recording and control interface with the vehicle, and CAN_to_ROS [14] provides a dynamic bridge between the vehicle network and the ROS [24] network. With these abstraction layers in place, the CBF-based control algorithm can be implemented in a Simulink model, and correct-by-construction ROS code can be generated to control the vehicle. Variations or updates to the software version can then be managed by an app-based scaffolding [8] which manages versioning, and defined switching between prescribed vehicle software applications. In Figure 2 the logic for how data and control commands flow is shown. In figure 1 a snapshot of the AV driving through traffic along the freeway corridor is shown.



Figure 1: View from inside the experimental vehicle during testing.

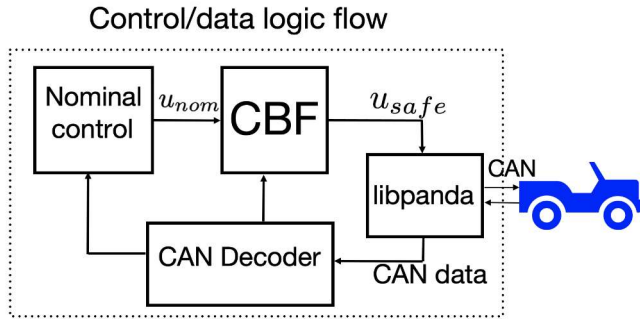


Figure 2: Logic flow for how control messages pass between different software interfaces is shown.

Control implementation. Via the CAN bus injection software/hardware interface control commands can be sent as well as system states measured from on-board sensors. Sensors measure vehicular speed ($v(t)$), vehicular acceleration ($\dot{v}(t)$), space-gap ($s(t)$), and relative speed difference ($\Delta v(t)$). Commanded acceleration ($u(t)$) is broadcast onto the CAN network, which low level engine control then responds to. Space-gap and relative speed difference are measured via front facing radar. The measurements of $v(t)$, $s(t)$, $\Delta v(t)$ are used to calculate control acceleration inputs in real time.

Nominal control is specified by a velocity control algorithm of the following form:

$$u_{nom}(\mathbf{x}) = k_{nom}(v - v_{des}) \quad (16)$$

where u_{nom} is the nominal acceleration control command, k_{nom} is a gain parameter, and v_{des} is the desired velocity. This nominal controller has no explicit safety considerations. For further details on the nominal control we direct the reader to [23].

The structure for the CBF deployed onto the AV is of the form:

$$u_{CBF} \leq \left(\frac{1}{t_{min}}\right)(k(s - (t_{min}v + d_{min})) + \Delta v) \quad (17)$$

where the choice of extended class- $\mathcal{K}$ function is

$$\alpha(h) = kh, k \in \mathcal{R}_{>0}^1$$

We use values of $t_{min} = 2.0[s]$, $d_{min} = 15.0[m]$, and $k = 0.1$. Equation (17) is the safety kernel for acceleration commands under

the minimum time-gap property and the choice of $\alpha(h) = kh$. For these values, forward-invariance of the safety property is as follows:

$$s(t_0) \geq 2.0v(t_0) + 15.0 \implies s(t) \geq 2.0v(t) + 15.0 \forall t \geq t_0$$

The final control acceleration command sent to the vehicle is:

$$u_{cmd} = \min(u_{CBF}, u_{nom})$$

where u_{Nom} is control specified by nominal control and u_{cmd} is the sent control command. This formulation is a specific solution to Equation (6) for this problem formulation.

Data processing. All sensor data over the course of the experiment is recorded and used to quantify behavior of the AV under CBF control. We accumulate recorded data into trajectories to analyze car-following behavior. A trajectory is defined as the set of states of the vehicle when following a single vehicle over a following period. We define a following period as the period in which the control AV is within 120[m] of another vehicle. If a merging event changes the leading vehicle or the leading vehicle exceeds the maximum following distance (120[m]) the trajectory ends. A new trajectory begins at the point that a new lead vehicle is within the maximum following distance. Additionally, for reasons further explained in Section 4, we sort trajectories into those that begin in satisfaction of the safety property (e.g. $h \geq 0$) and those that begin in violation of the safety property (e.g. $h < 0$).

We also note that at the point of a merge-in event, the environmental signal $\mathbf{e}$ is discontinuous, meaning $\dot{\mathbf{e}}$ is not defined at that point. After the merge-in is complete we assume that $\mathbf{e}$ can be considered Lipschitz and continuous. As a result when engaged in uninterrupted car-following (e.g. no merges) the application of environmental CBFs is a valid approach for safety filtering. We focus on the car-following trajectories where this is the case.

Across the experiment 25 different trajectories were recorded. The majority of trajectories are of speeds under $15[\frac{m}{s}]$ and space-gaps under $80[m]$. This is consistent with driving in congested traffic. In total around 1.17 hours of car-following were collected over the experiment. 11 trajectories begin in violation and 14 begin in satisfaction of the safety property.

4 RESULTS

In this section the results of the experimentation are presented. We present results characterizing the extent to which the CBF controlled AV satisfies the associated safety properties. We show an analysis of forward-invariance of the safety property. We then analyze recovery when a trajectory begins in violation of the of the safety property. Finally, we analyze errors between control inputs and achieved outputs and how these are related to the CBF performance.

4.1 CBF performance

Here we summarize the general performance of the CBF-enabled AV.

In Figure 4 the cumulative and probability distributions of all values of h across all trajectories are shown. A significant portion of h values are centered around 0, but significant outliers exist which indicate large violations to the safety property at some times during the experiment.

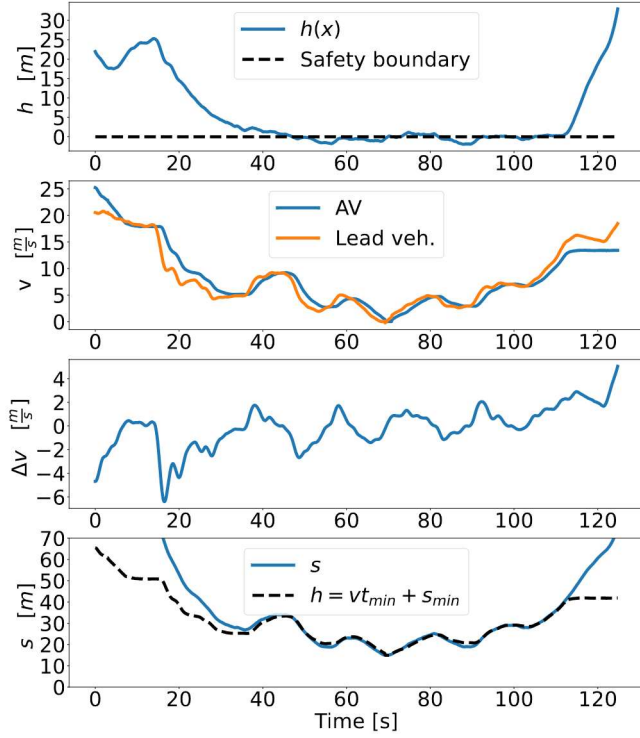


Figure 3: An example of data collected from a single trajectory is shown. h, s, v and the leading vehicular speed ($v + \Delta v$) are shown over the course of the trajectory

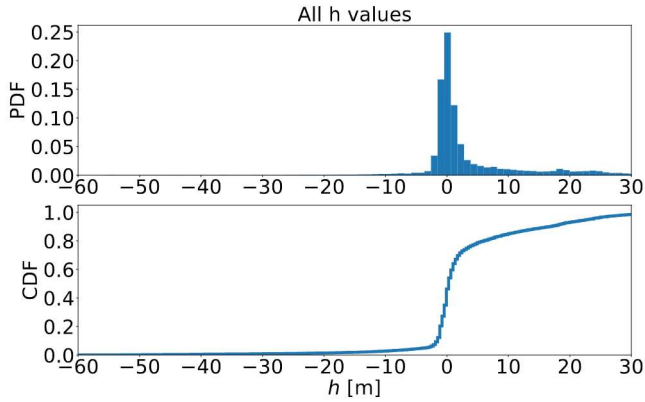


Figure 4: Cumulative and probability distributions of values of h across the entire experiment are shown.

In Figure 5 the distribution (both density and cumulative) of all measured space gap values are shown. A significant portion (around 80%) of the measured space gaps are under 40 [m]. The minimum measured space-gap value across all trajectories was 11.7 [m]. Despite violations to the safety property, space-gap values remain far above 0, meaning that the AV never came close to a collision.

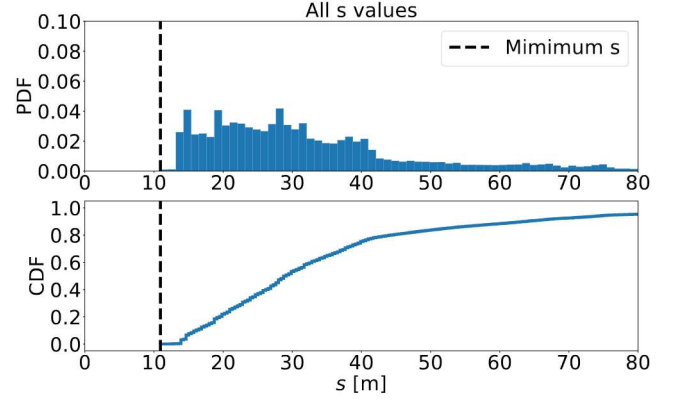


Figure 5: Cumulative and probability distributions of space-gap across the entire experiment are shown. The minimum observed space-gap of 11.7[m] is shown in black.

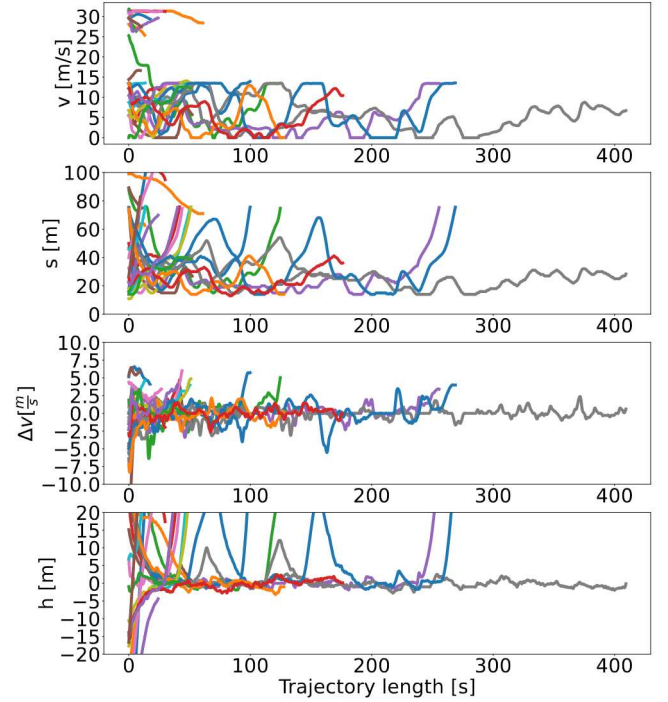


Figure 6: All trajectories across the experiment are plotted, with speed (v), space-gap (s), speed difference (Δv) and h shown. Trajectories are plotted over time starting from the beginning of each trajectory.

All trajectories across the entire experiment are shown in Figure 6, where speed, space-gap, speed difference, and h are all plotted over time from the beginning of the trajectory. Trajectories start both within the safe set and outside. These results motivate the need to explore both forward-invariance and recovery respectively.

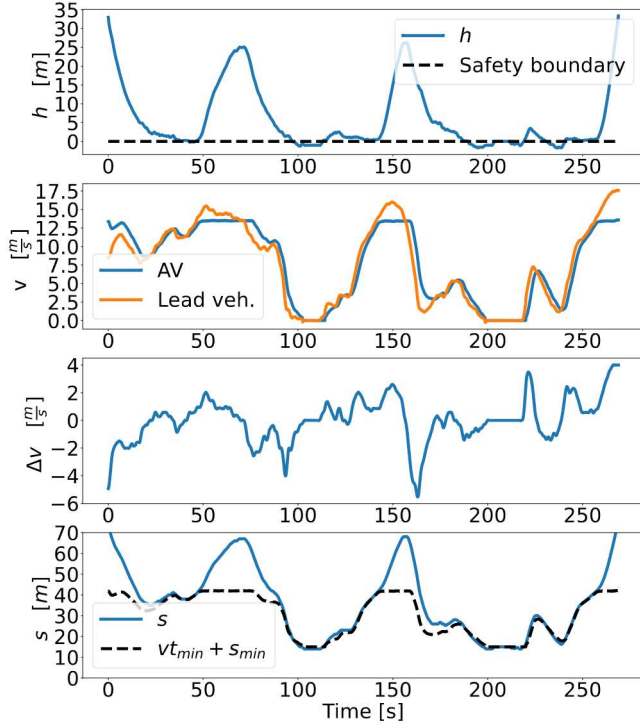


Figure 7: An example of a trajectory that begins satisfying the safety property is shown. h , speed, speed difference, and space-gap are all plotted over the duration of the trajectory.

4.2 Safety invariance

Here we consider trajectories that begin in satisfaction of the safety property. For a trajectory to be forward-invariant with respect to the safety property it must first begin inside the safe set of states. Here we analyze the extent to which those trajectories which should be forward-invariant (e.g. start inside the safe set) are, or are not. The main findings are that trajectories beginning in satisfaction of the safety property do not always remain inside the safe set. This means the AV is not strictly forward-invariant in the safety property. However, the degree of violation to the safety property is under $3.4[m]$ and mostly under $2[m]$.

In Figure 7 an example of a trajectory beginning inside the safe set is shown. As the AV approaches the lead vehicle the safety boundary is approached and subsequently crossed by an amount under $2[m]$. The safety boundary is at $h = 0$. In the top subplot the evolution of h over time is shown. In the bottom subplot the value of $vt_{min} + s_{min}$, along with spacing s , is shown over time. Periods where the $s < vt_{min} + s_{min}$ correspond to $h < 0$. In the middle two subplots the speed (v) of the AV and lead vehicle are shown as well as the corresponding speed difference (Δv).

Across all trajectories beginning in the safe set we quantify subsequent violations to the safety property. In Figure 8 the distribution of all h values for trajectories initially in the safe set are shown. Both the probability distribution function (PDF) and cumulative distribution function (CDF) are shown. The distributions show that

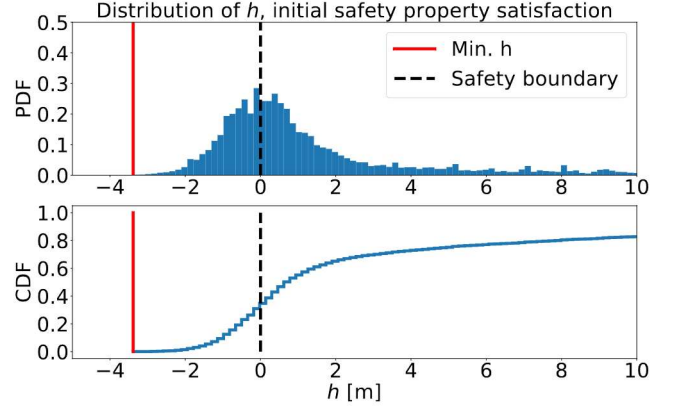


Figure 8: Probability and cumulative distributions of all measured values of h for trajectories beginning in satisfaction of the time-gap safety property are shown. The minimum h value of -3.4 from this set is shown in red.

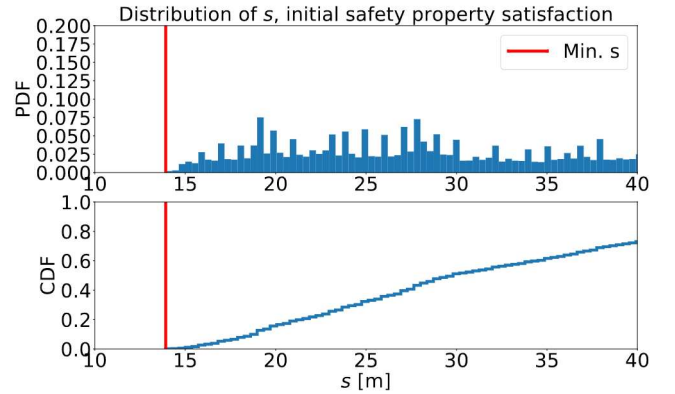


Figure 9: Probability and cumulative distributions of all measured values of s for trajectories beginning in satisfaction of the time-gap safety property. The minimum value from this set of $13.9[m]$ is shown in red.

approximately 40% of all h when initially in the safe set are subsequently outside of the safe set, showing that forward-invariance of the safety property is not strictly achieved. In red the minimum value of h is shown, which is $-3.4[m]$. This means that the largest violation to the safety property was by $3.4[m]$. Additionally, over 97% of all h values when initially in the safe set were such that $h > -2.0$.

The distribution of space-gap values from this set of trajectories is shown in Figure 9. The probability and cumulative distributions are shown. The minimum space-gap from this set of trajectories was $13.9[m]$. Across these trajectories the AV remained far from collisions.

The evolution for all trajectories beginning the safe set is shown in Figure 10. Each trajectory is plotted in the $s-v$ plane and colored with respect to $\dot{h}$. The safety boundary ($h = 0$) is displayed as dashed

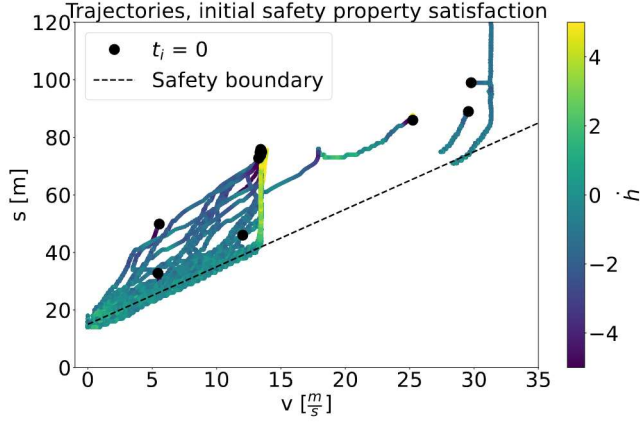


Figure 10: Trajectories with initial satisfaction of the time-gap safety property are shown in $s-v$ and colored with respect to $\dot{h}$.

black. The beginning point of the each trajectory is shown as a black dot.

The primary conclusions from analyzing trajectories beginning in initial safety are as follows: Forward-invariance of the safety property does not strictly hold. The maximum violation to the safety property was 3.7[m]. The AV remained above 13.9[m] spacing across these trajectories.

4.3 Safety recovery

Here we discuss the recovery of the CBF controlled AV for trajectories that begin in violation of the safety property. To reiterate, recovery refers to the property $h < 0 \implies \dot{h} > 0$. We quantify the extent to which this property holds by analyzing all trajectories which begin outside of the safe set. This happens when a lead vehicle merges in front of the AV at a distance such that $h < 0$. Our primary findings are as follows: Recovery did not strictly hold, there exist measured states at which $h < 0$ and $\dot{h} < 0$. As h is more negative (more of a violation to the safety property) the percentage of states that are recovering increases. For $h < -5$ 95% of measured states were recovering.

An example trajectory that begins outside the safe set is shown in Figure 11. In the top subplot values of h are shown over the duration of the trajectory. The second subplot shows the speeds of the AV and the leading vehicle. The third subplot displays both the relative speed difference (Δv) and $\dot{h}$ with respect to the recovery boundary $\dot{h} = 0$. In the fourth subplot the space gap and $vt_{min} + s_{min}$ are shown. We calculate $\dot{h}$ as $\dot{h} = \dot{v}t_{min} + \Delta v$ using the measured acceleration rather than the commanded acceleration. This reflects the measured value of $\dot{h}$ rather than the value according to the modeling assumption in Equation (10).

The trajectory begins outside of the safe set, e.g. $h < 0$. Over time the trajectory reaches the boundary of safety. At the beginning of the trajectory the AV is initially not recovering. Subsequently $\dot{h} > 0$ for the next 20 seconds as safety boundary is approached. This trajectory shows an example of how recovery is not strictly

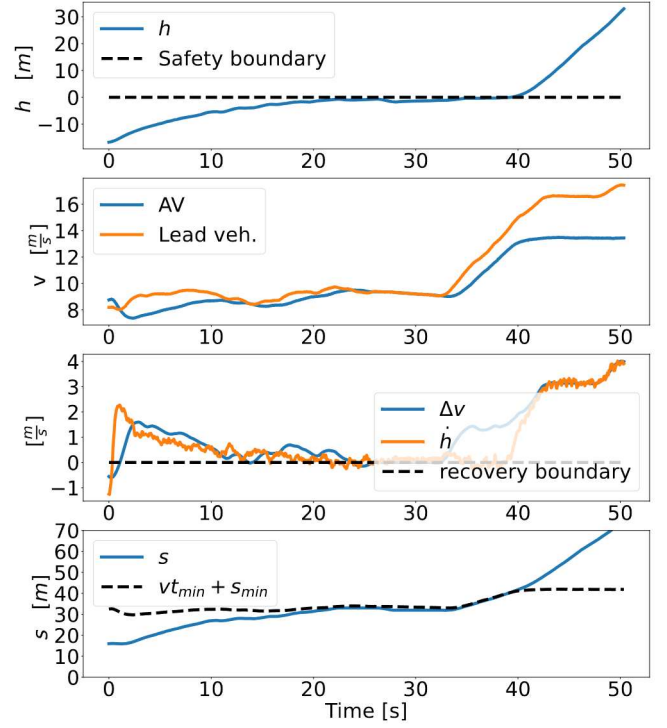


Figure 11: An example of a trajectory beginning in violation of the safety property that moves back to the safe boundary. h , speed, speed difference and $\dot{h}$ are all shown over the duration of the trajectory.

followed, but that the trajectory approaches back to the safety boundary over time and that for much of this period $\dot{h} > 0$.

An example of a trajectory beginning outside the safe set is shown in Figure 11. h is shown in the first subplot, speed in the second, both speed difference and $\dot{h}$ in the third, and space-gap in the final. Over the course of the trajectory the AV eventually returns to the safety boundary at $h = 0$. For a period at the beginning of the trajectory $\dot{h} < 0$, which then becomes positive as the AV returns to the safety boundary.

In Figure 12 the distribution of $\dot{h}$ when $h < 0$ is shown across all trajectories. The existence of $h < 0$ and $\dot{h} < 0$ shows further that recovery did not strictly hold across all states.

In Figure 13 the distribution of $\dot{h}$ when $h < 0$ is shown in more detail. The cumulative distribution is shown for $\dot{h}$ for $h < \delta$ where δ are increasing amounts of violation to the safety property. At $\delta = 0.0$ close to 40% of $\dot{h}$ values are < 0 . In comparison, at $\delta = -10.0$ fewer than 2% of $\dot{h}$ values are < 0 . This shows that as the AV is further from the safety boundary it is recovering a higher percentage of the time.

Figure 14 shows all points with $h < 0$ in terms of $\dot{h}$ vs. h . In keeping with the trend shown in Figure 13 as $h \ll 0$ the AV is mostly recovering, $\dot{h} > 0$, but not at all points.

In Figure 15 the state of the vehicle is shown in the $s-v$ plane and colored with respect to $\dot{h}$ for each trajectory beginning outside of the safe set. Generally, as the state of the vehicle is further from the

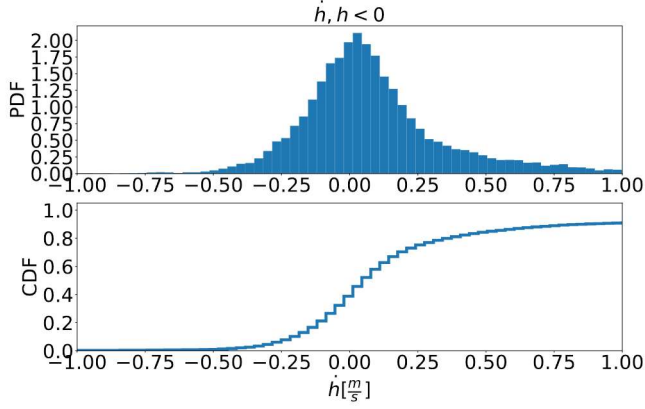


Figure 12: Probability and cumulative distributions of $\dot{h}$ when $h < 0$ are shown.

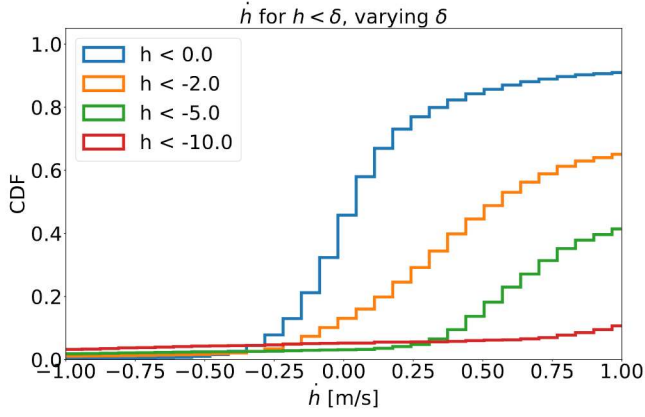


Figure 13: $\dot{h}$ dot varying degree of violation

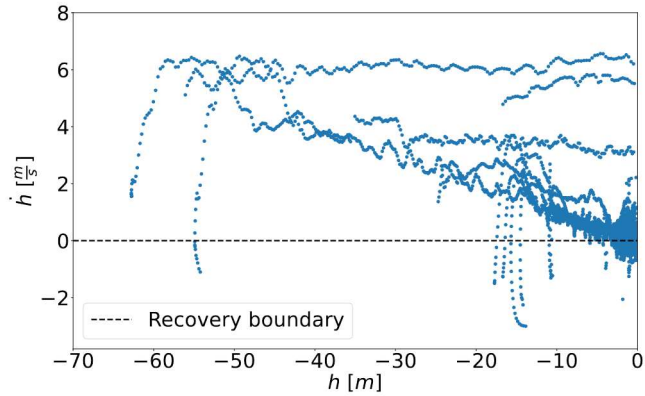


Figure 14: $\dot{h}$ is plotted against h for all $h < 0$. As $h \ll 0$ mostly $\dot{h} > 0$ with some exceptions.

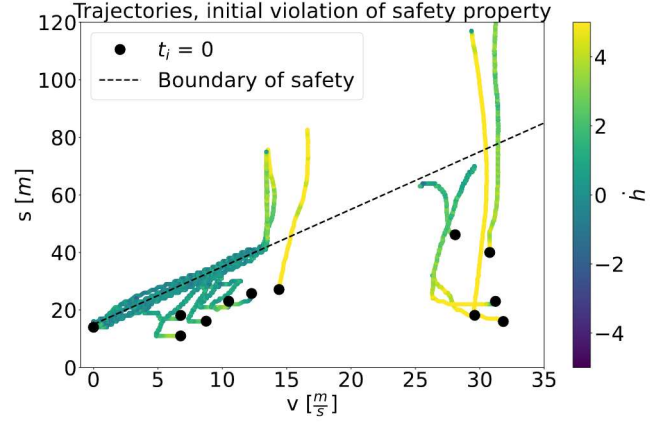


Figure 15: Trajectories starting in violation are shown in the $s-v$ plane and colored with respect to $\dot{h}$. Initial states are denoted as black dots.

safety boundary (e.g. $h \ll 0$) $\dot{h}$ is more positive. This corresponds with faster recovery further away from the boundary of the safety property.

The core findings from this analysis are as follows: the recovery property is not strictly met. For h further from the safety boundary a higher percentage of $\dot{h}$ are positive. Over time trajectories beginning outside the safe set approach back towards the boundary of safety.

4.4 Analysis of unaccounted model error

Given that both the forward-invariance property and the recovery property are not strictly met, errors between the assumed model in Equation (10) and the real world behavior of the AV are present. Here we analyze different errors between expected quantities and what was achieved in testing.

In the model introduced in Equation (10) it assumed that the acceleration of the AV can be directly controlled. To quantify the extent to which this is an accurate modeling assumption we look at $u - \dot{v}$, which is error on acceleration commands compared to achieved acceleration. In Figure 16 the distribution of $u - \dot{v}$ across all states is given. This shows that $u \neq \dot{v}$ at all times, meaning that the assumed model is not strictly correct. The mean of this distribution is $-.031 \frac{m}{s^2}$ with a standard deviation of $.22 \frac{m}{s^2}$. Across all states 78% were such that $u - \dot{v} < 0$. From the error distribution in Figure 16 it can be concluded that the assumed model is not strictly correct, there are observed errors between commanded acceleration and achieved acceleration.

This trend is also present when looking at states which are in violation of the safety property. Figure 17 shows the distribution of all $u - \dot{v}$ across states which are in violation of the safety property. A similar distribution to that in Figure 16 is present. The mean of this distribution is $-.038 \frac{m}{s^2}$ with a standard deviation of $.17 \frac{m}{s^2}$. 77% of measured states were such that $u - \dot{v} < 0$.

For $u < 0$, then $u - \dot{v} < 0$ corresponds to the AV under-braking compared to what the CBF logic commands. We hypothesize that this under-braking due to un-accounted for model features could be a contributing factor to violations on the safety property.

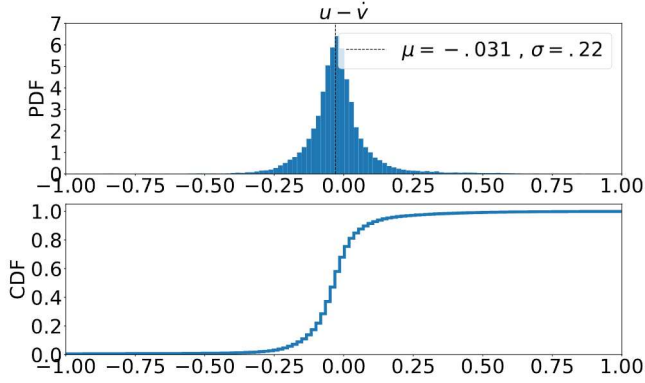


Figure 16: The distribution of error on commanded control compared to achieved acceleration are shown.

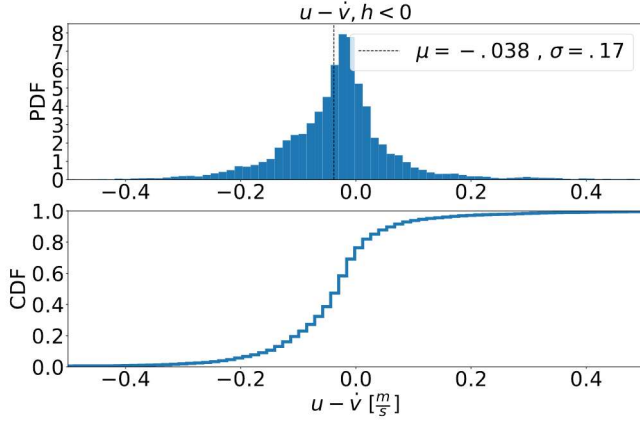


Figure 17: The distribution $u - \dot{v}$ when $h < 0$.

Figure 18 shows part of the trajectory that has the largest violation to forward-invariance of all trajectories beginning in the safe set. The top subplot shows the value of h . At 122 seconds into the trajectory $h = -3.4[m]$. In the second subplot the AV's speed v and the leading vehicles speed v_l are shown. Over this portion of the trajectory the lead vehicle accelerates and then decelerates coming to a full stop. In the fourth subplot the commanded acceleration u and achieved acceleration $\dot{v}$ are shown. The CBF control logic first commands acceleration as the lead vehicle speeds up, and subsequently braking as the lead vehicle slows. The acceleration error $u - \dot{v}$ as well as $\alpha(h) + \dot{h}$ are shown in the third subplot. In the period from 90 – 130[s] $u - \dot{v}$ is predominantly negative. As the CBF commands braking, the AV's achieved acceleration is above the commanded acceleration, corresponding to under-braking. During this time $\alpha(h) + \dot{h}$ is also predominantly negative, and it is over this period that the safety property is violated.

The primary results from this analysis are as follows: Errors between commanded acceleration and achieved acceleration are present meaning the assumed model in Equation (10) is not strictly correct in practice. The achieved acceleration values tend to be more positive than the commanded accelerations. When braking is

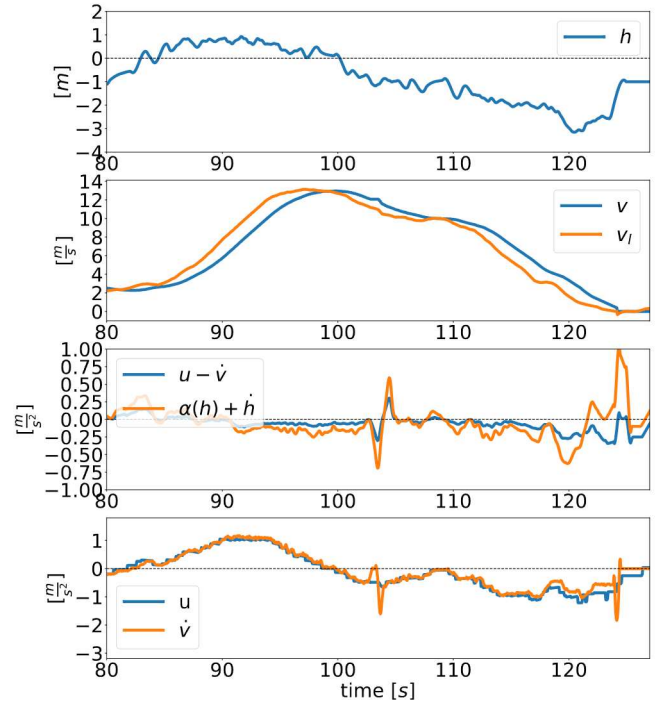


Figure 18: The trajectory with the largest degree of violation to the time-gap safety property when beginning in satisfaction is shown.

commanded this corresponds with under-braking. We hypothesize that this could be a significant contributing factor to violations forward-invariance and recovery.

5 CONCLUSIONS AND DISCUSSION

Conclusions. This work presents experimental results from testing an automated vehicle supervised for safety by a control barrier function in live freeway traffic. We look for the extent to which the following properties were met: forward-invariance of a minimum time-gap safety property, recovery of that safety property when in violation, and collision avoidance.

To analyze forward-invariance we examine trajectories of continuous car-following that begin with the safety property satisfied. We find that over the course of the experiment these trajectories are not strictly forward-invariant, the AV leaves the set of safe states. The maximum amount the safety property was violated by was $3.4[m]$. More than 95% of states measured in these trajectories were such that $h \geq -2.0$.

We analyze the recovery of the CBF controlled AV by looking at trajectories which begin outside the safe set of states. This is caused by a merge-in event at a distance closer than the minimum safe distance. Similarly to forward-invariance we find that the recovery property is not strictly met. There were measured states such that $h < 0 \wedge \dot{h} < 0$. Across states such that $h < -5$ it was found that the recovery property was met more than 95% of the time.

Finally, across all trajectories we found that the minimum spacing-gap observed was 11.7[m]. This result suggests that while the minimum time-gap safety property was not strictly forward-invariant nor strictly recovering in violation, the CBF achieved good performance in terms of preventing collisions. These results together suggest that control barrier functions are a viable approach for managing safety in the context of AVs.

Discussion. A primary consideration for future work is how changes to modeling assumptions can change the CBFs structure and their performance. Since the original introduction of control barrier functions for control of AVs [4] a number of works have demonstrated how more complicated models could be considered, including those with input delay [21], adaptive models [31], or deep-learning based models [26]. Continued experimental testing of these different approaches is needed to understand which theoretical approaches translate to real world deployment.

In future work we will also investigate how different nominal control algorithms affect the overall safety of the system. It is possible that other nominal control algorithms could place the AV in situations in which the CBF is either better suited or worse suited to supervise safety than the nominal controller used in these experiments.

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